virus will encrypt certain files on the hard drive and any mounted storage connected to it with RSA public key cryptography. The original ransomware on the market was CryptoLocker, and the hackers would demand a ransom (hence, the name) in exchange for the decryption key. In June 2014, Operation Tovar took down Evgeniy Bogachev, the leader of the gang of hackers behind CryptoLocker, but many knockoffs are still running around in the market, though the affected user base is a much smaller one. CryptoLocker managed to affect around 500,000 users in its 100 days, and the hackers made off with upwards of $30 million with this heist.

Keyloggers and Viruses

Then there are the multitude of viruses and malware that steal passwords and user account details by logging in keyboard strokes whenever the user visits any website. There's the Gameover ZeuS trojan, which steals one's login details on popular Web sites that involve monetary transactions. It works by detecting a login page, then proceeds to inject a malicious code into the page, keystroke logging the computer user's details. Zeus was been created to steal private data from the infected systems, it's still customisable to gather banking details in specific countries and by using various methods.

But the worst of viruses, which is the first example of the potential of cyber-terrorism, is the Stuxnet computer worm, believed to have been created to sabotage Iran's nuclear program. The virus is typically introduced to the target environment via an infected USB flash drive, which then introduces the infected root-kit into the system, modifying the codes and giving unexpected commands to the computer while returning a loop of normal operations system values feedback to the users. The aim of the worm was to fake industrial process control sensor signals so that the infected system does not shut down due to detected abnormal behaviour.

While most attacks have been limited to computing systems like servers, desktops and laptops, hackers now focus on the IoT ecosystem and all its connected components. Large networks of IoT devices—like CCTV surveillance cameras, smart TVs, and home automation systems are prone to hacking, and the modern age thief will make use of these to carry out coordinated attacks against individuals and corporations. So unless we beef up our arsenal against these goons, they'll move on to stealing biometric data and personal information that can be used to impersonate fully functioning individuals. d